FaceTrack

User Guide

A complete, step-by-step walkthrough of the Face Recognition Based Classroom Attendance System — every screen, every user flow, and the accounts you need to reach each part of the platform.

LIVE APPLICATION

https://attendace-system-david.vercel.app/

Document	System	Audience	Date
User Guide v1.0	**FaceTrack Attendance**	**Administrators & Lecturers**	**September 2026**

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How to read this guide

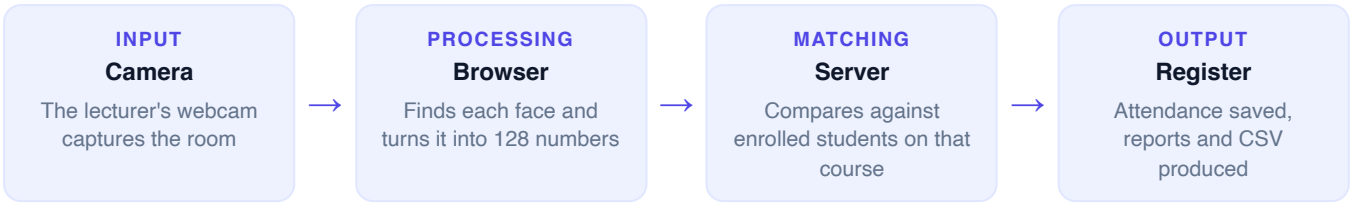
Sections 1–3 get you signed in. Section 4 explains every screen you will meet. Section 5 strings those screens into the three journeys you will actually perform. If you are demonstrating the system and only have five minutes, go straight to the quick reference card in section 8.

1. What FaceTrack does

Attendance that is taken by the camera, not by a register passed around the hall.

In most lecture halls attendance is still taken on paper. It eats lecture time, the sheets get lost, and students sign for friends who are not in the room. FaceTrack replaces that with a camera at the front of the class: a student looks at it, is recognised from their own face, and is marked present in under a second.

How the parts fit together



Your camera footage never leaves your computer

The browser converts each face into a list of 128 numbers — a "descriptor" — and only those numbers are sent to the server. No photograph of the classroom is ever uploaded, and a descriptor cannot be turned back into a picture of a face.

What it gives you

CAPABILITY	WHAT IT MEANS IN PRACTICE
Instant attendance	No roll call. The camera scans continuously while you teach.
No proxy attendance	A student must present their own face. Naming somebody is not enough — matching happens on the server, against templates the browser never sees.
No double entries	A student who stays in front of the camera for ten minutes still produces exactly one record. The database itself enforces this.
Late tracking	Anyone recognised after your cut-off (15 minutes by default) is marked *late* rather than *present*.
Locked registers	Closing a session freezes it. Attendance cannot be added to a lecture that has finished.
Reports & eligibility	Attendance percentage per student, flagged against the course requirement (75% by default), exportable as CSV.

2. Before you start

Three things to check before your first lecture.

2.1 What you need

REQUIREMENT	DETAIL
Browser	Google Chrome, Microsoft Edge or Safari, kept up to date. Chrome gives the smoothest camera performance.
Camera	Any built-in laptop webcam or USB camera. A camera you can angle towards the students works best.
Internet	Needed to reach the site and save attendance. Face detection itself runs on your own machine.
Lighting	Faces must be lit from the front. Students backlit by a window are the single most common cause of a missed face.

2.2 The first load takes a moment

The first time you open the **Register student** or **live capture** screen in a given browser, roughly **12 MB** of face-recognition models download in the background. The camera button stays greyed out and reads `Loading face detector…` until this finishes — usually 10 to 40 seconds. The browser caches them afterwards, so it only happens once.

> **Open the capture screen a minute before your lecture starts**
> That lets the models finish downloading while students are still settling, instead of during the first minute of class.

2.3 The camera only works on a secure address

Browsers refuse camera access unless the page is on `https://` or `localhost`. The live site is on HTTPS, so it works:

ADDRESS	CAMERA
`https://attendace-system-david.vercel.app/`	✓ Works
`http://localhost:3000` (running it yourself)	✓ Works
`http://192.168.x.x:3000` (a LAN address)	✗ Blocked by the browser

3. Accounts & passwords

Students never sign in — they are identified by their face. Only staff have accounts.

3.1 Sign-in details

Open https://attendace-system-david.vercel.app/ and you will land on the sign-in screen. Use one of these accounts:

Administrator — full access

Email	`admin@example.edu`
Password	`admin1234`
Signs in as	**Dr. Grace Adeyemi**

Use this account to **register students, capture face templates, create courses and manage staff accounts**. It sees every course in the system.

Lecturer — teaching access

Email	`lecturer@example.edu`
Password	`lecturer1234`
Signs in as	**Mr. Emeka Nwosu**

Use this account to **run lectures** — start sessions, capture attendance, close registers and read reports. It only sees the courses assigned to it.

These are demonstration passwords

They are deliberately simple so the system can be shown and marked. Before the platform is used with real student data, create fresh accounts under **Staff accounts** with strong passwords and delete these two.

3.2 What each role can do

ACTION	ADMINISTRATOR	LECTURER
Sign in and see the dashboard	✓	✓
Register a new student and capture their face	✓	✗
Edit, re-enrol or delete a student	✓	✗
Create or delete a course	✓	✗
Add students to a course class list	✓	✓
Start a lecture session	✓	✓
Capture attendance with the camera	✓	✓
Mark a student manually	✓	✓
Close or reopen a session	✓	✓
Remove a wrong attendance record	✓	✓
View reports and export CSV	✓	✓
See *every* course in the institution	✓	✗ own only
Create staff accounts	✓	✗

Signing out

Your name sits at the bottom of the left sidebar with a `Sign out` button beneath it. Sessions expire automatically after 8 hours.

4. The screens, one by one

Every screen in the platform, what it is for, and exactly what to click.

4.1 Sign in EVERYONE `/login`

The entry point. There is no student login and no public sign-up — accounts are created by an administrator.

1. Open `https://attendace-system-david.vercel.app/`. You are redirected here automatically.
2. Enter your **email address** and **password** from section 3.1.
3. Press **Sign in**. You land on the Dashboard.

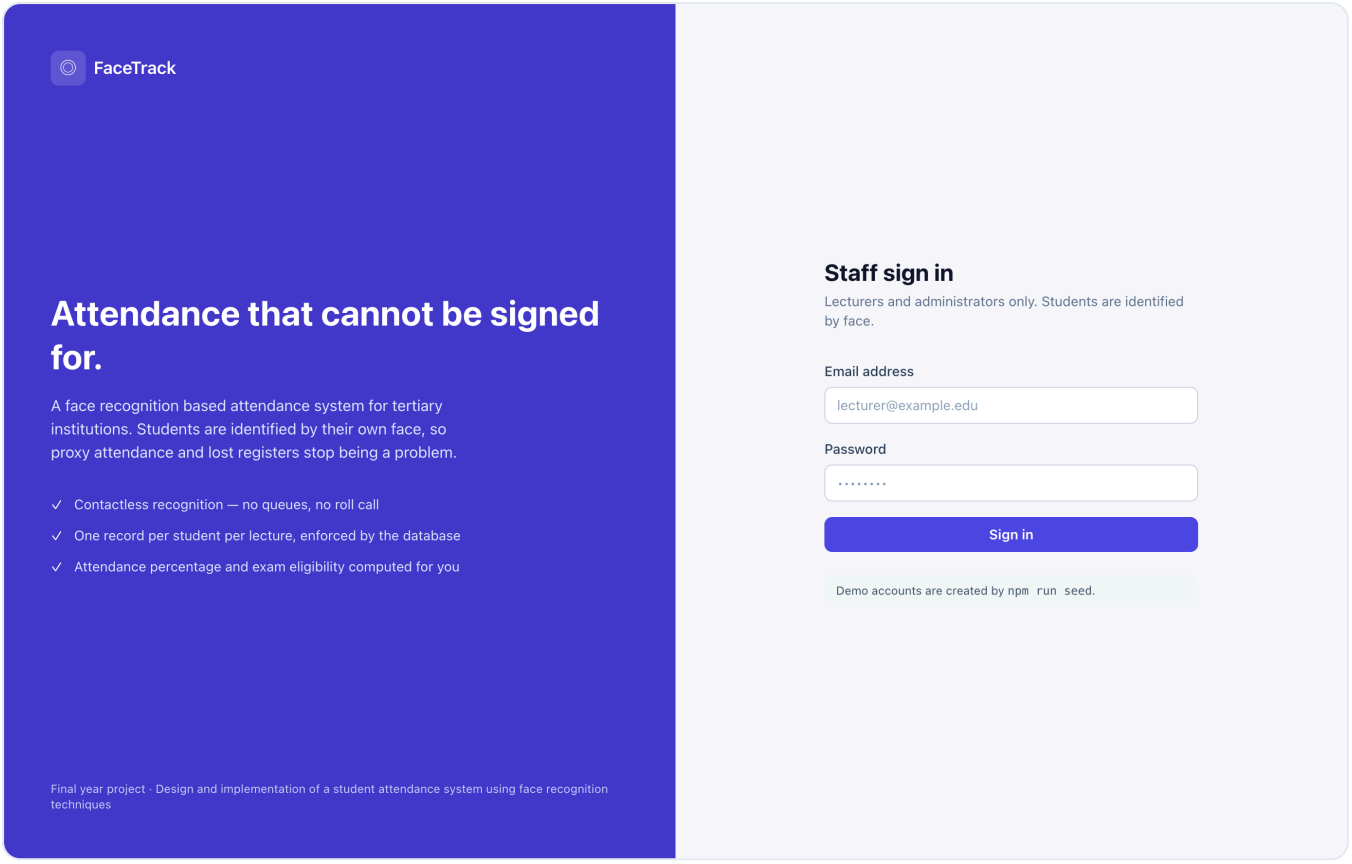


Figure 1. The sign-in screen. Wrong details produce a red message under the password field — the system never reveals whether it was the email or the password that was wrong.

4.2 Dashboard EVERYONE `/dashboard`

Your landing page and the fastest route back into a lecture that is already running.

- **Registered students** — total on the system, with a warning chip counting those who still have no face template.
- **Courses** — how many courses you are responsible for.
- **Open sessions** — lectures currently accepting attendance.
- **Marked today** — records saved today, split between face and manual.

- **Open lecture sessions** — each has an **Open camera** button that takes you straight to capture.
- **Recent attendance** — the last eight students marked anywhere in the system.

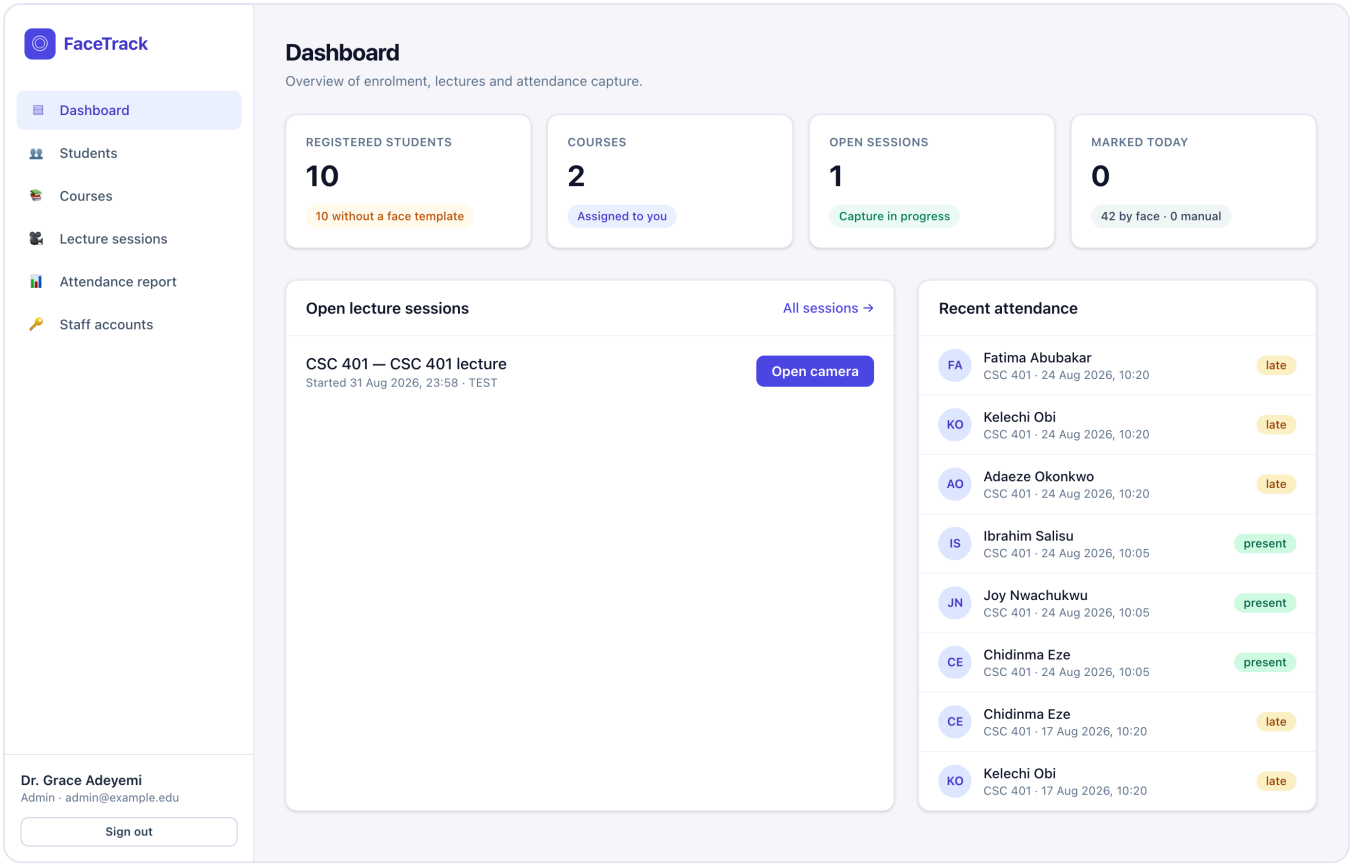


Figure 2. The dashboard. The amber chip — *"10 without a face template"* — is the one to watch: those students cannot be recognised until somebody enrols their face.

4.3 Students EVERYONE `/students`

The full student roll, with each student's biometric enrolment status.

1. Use the **search box** to filter by name, matriculation number or department as you type.
2. Read the **Face template** column: a green *"3 samples"* badge means the student can be recognised; a red *"Not enrolled"* badge means they cannot.
3. Click any student's name to open their profile.
4. Click **+ Register student** (top right) to add somebody new.

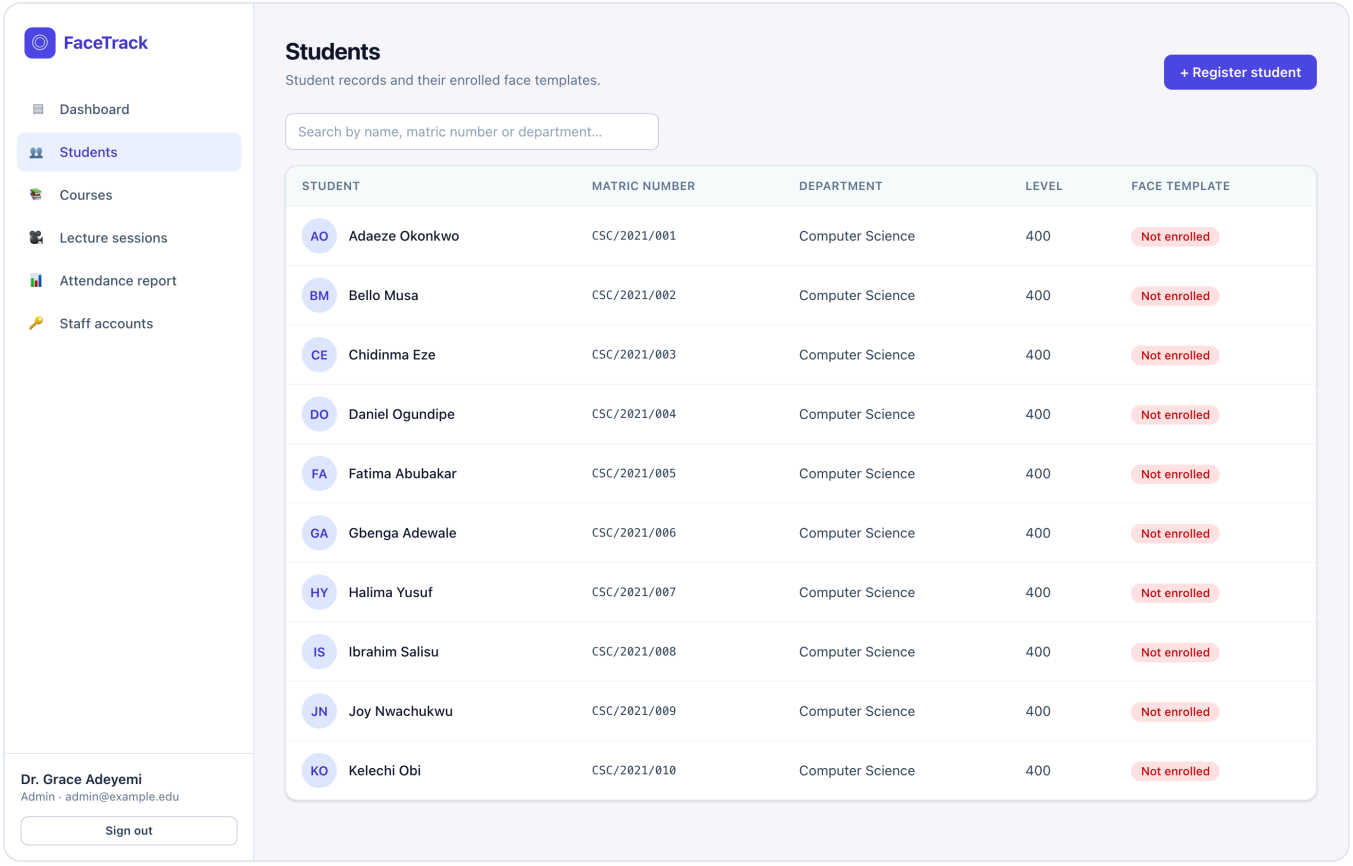


Figure 3. The student roll. Everyone here still shows *Not enrolled* — seeded records deliberately ship without face templates, because a template is only meaningful if it came from a real face.

4.4 Register a student & enrol their face ADMIN ONLY `/students/new`

This is the most important screen in the system. A student who has never been enrolled here can never be recognised in class.

1. Fill in the **matriculation number** and **full name** (both required). Email, department and level are optional but make reports easier to read.
2. On the right, wait for the button to change from *Loading face detector…* to **Turn on camera**, then press it. Allow camera access when the browser asks.
3. Sit the student squarely in frame, well lit, about an arm's length from the camera.
4. Press **Capture sample** three times, following the prompt under the video each time:
   - Sample 1 — *look straight at the camera*
   - Sample 2 — *turn your head slightly to the left*
   - Sample 3 — *turn your head slightly to the right*
5. Each successful capture flashes a green box and adds a thumbnail below. The counter climbs to **3/3**.
6. Press **Save student**.

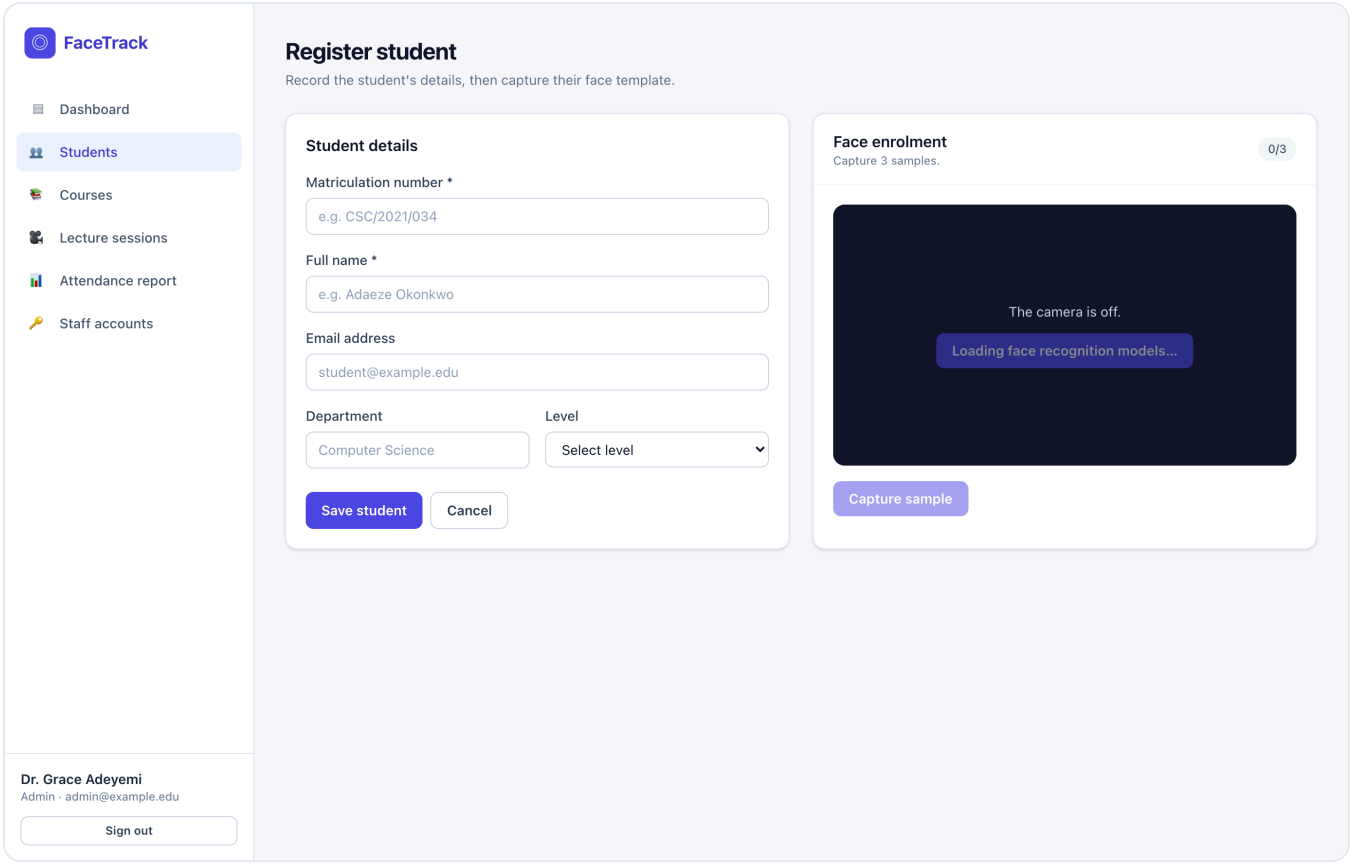


Figure 4. Registration. Details on the left, face enrolment on the right. The **0/3** counter tracks captured samples.

> **Why three samples from three angles?**
> A single head-on template fails the moment a student tilts their head in a real lecture. Three angles give the matcher room to recognise the same person in a natural pose — this is the single biggest thing you can do to improve accuracy in the hall.

> **If capture keeps failing**
> “No face detected” almost always means the light is behind the student. Turn them to face a window or a lamp. “Face is unclear” means they are too far away or moving.

4.5 Student profile EVERYONE `/students/[id]`

Everything recorded about one student.

- **Left panel** — photo, details, template status, and the courses they are registered on.
- **Attendance history** — the last 50 records with course, session, time, status and whether it was face or manual.
- **Re-enrol face** — captures three fresh samples and *replaces* the stored ones. Use this when a student is repeatedly not recognised, or their appearance has changed.
- **Delete** — removes the student, their face template and all their attendance. Admin only, and it cannot be undone.

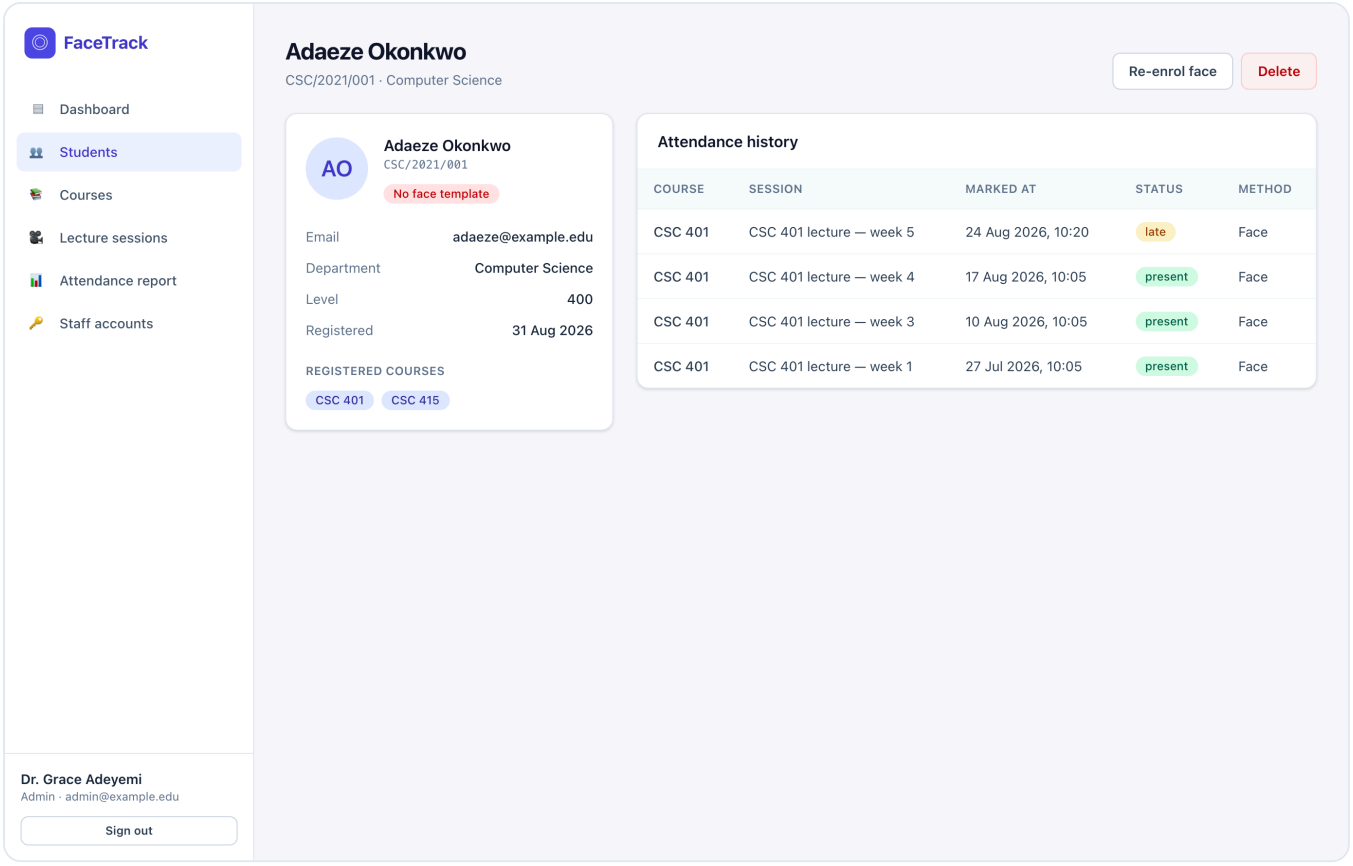


Figure 5. A student profile showing details, registered courses and attendance history.

4.6 Courses EVERYONE

`/courses`

Administrators see every course; lecturers see only their own.

1. Press **+ New course** to open the creation form.
2. Enter the **course code** and **title** (required), then level, credit unit, department and the **lecturer** who will teach it.
3. Set **Required attendance (%)** — the threshold used to decide exam eligibility. The default is 75.
4. Press **Create course**, then click the new card to add students to it.

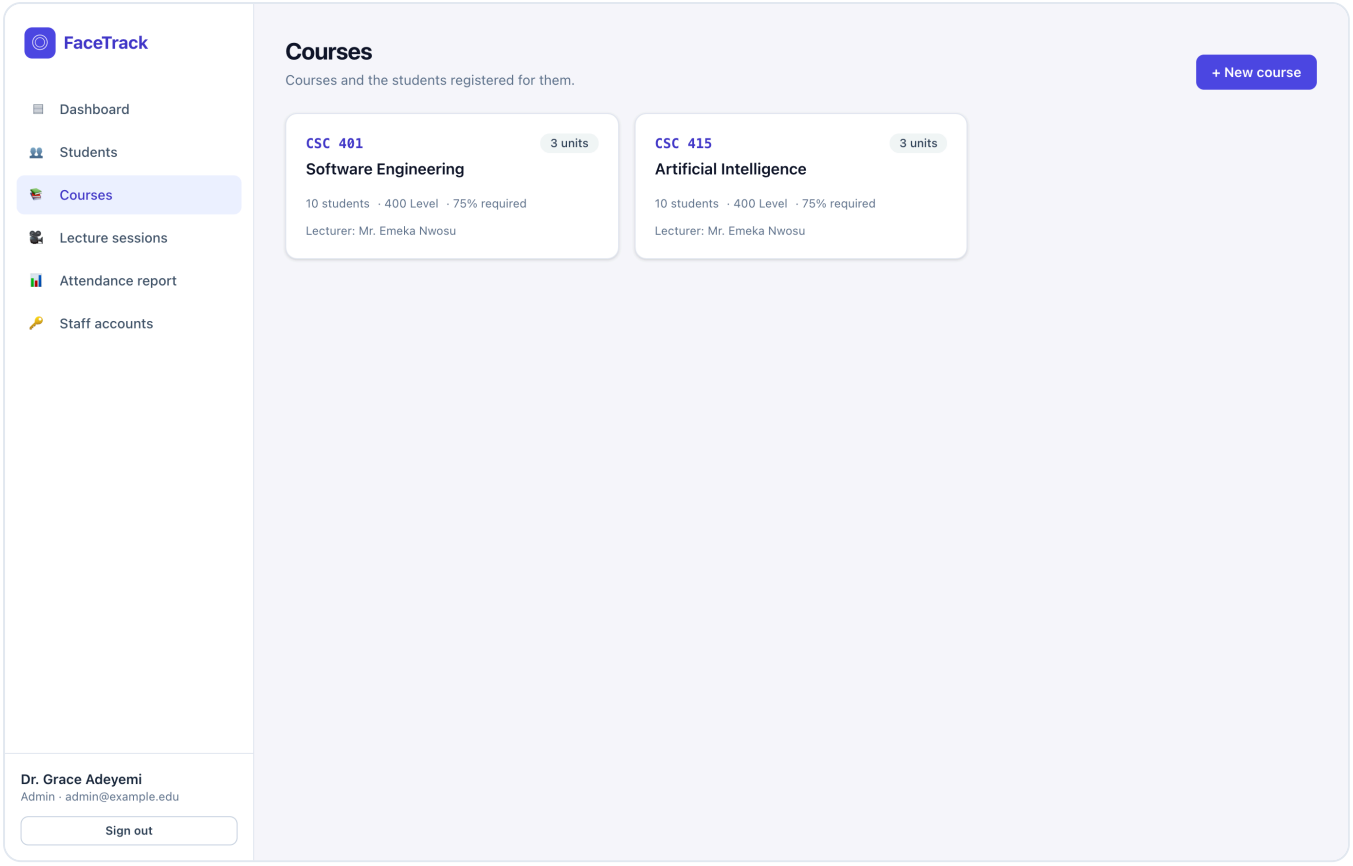


Figure 6. Course cards showing class size, level and the attendance requirement.

4.7 Course detail & class list EVERYONE `/courses/[id]`

The class list matters more than it looks: **only students registered on a course are candidates for recognition in that course's lectures.** This is what stops a student from another class walking past the camera and being marked present.

1. Press **+ Add students** to open the picker.
2. Search, then tick each student you want. A red *no face* badge warns you that a student has no template and will not be recognised.
3. Press **Register N student(s)**, then **Done**.
4. Use **Remove** beside any name to take them off the course.
5. **Start lecture session** (top right) begins a sitting and jumps straight to the camera.

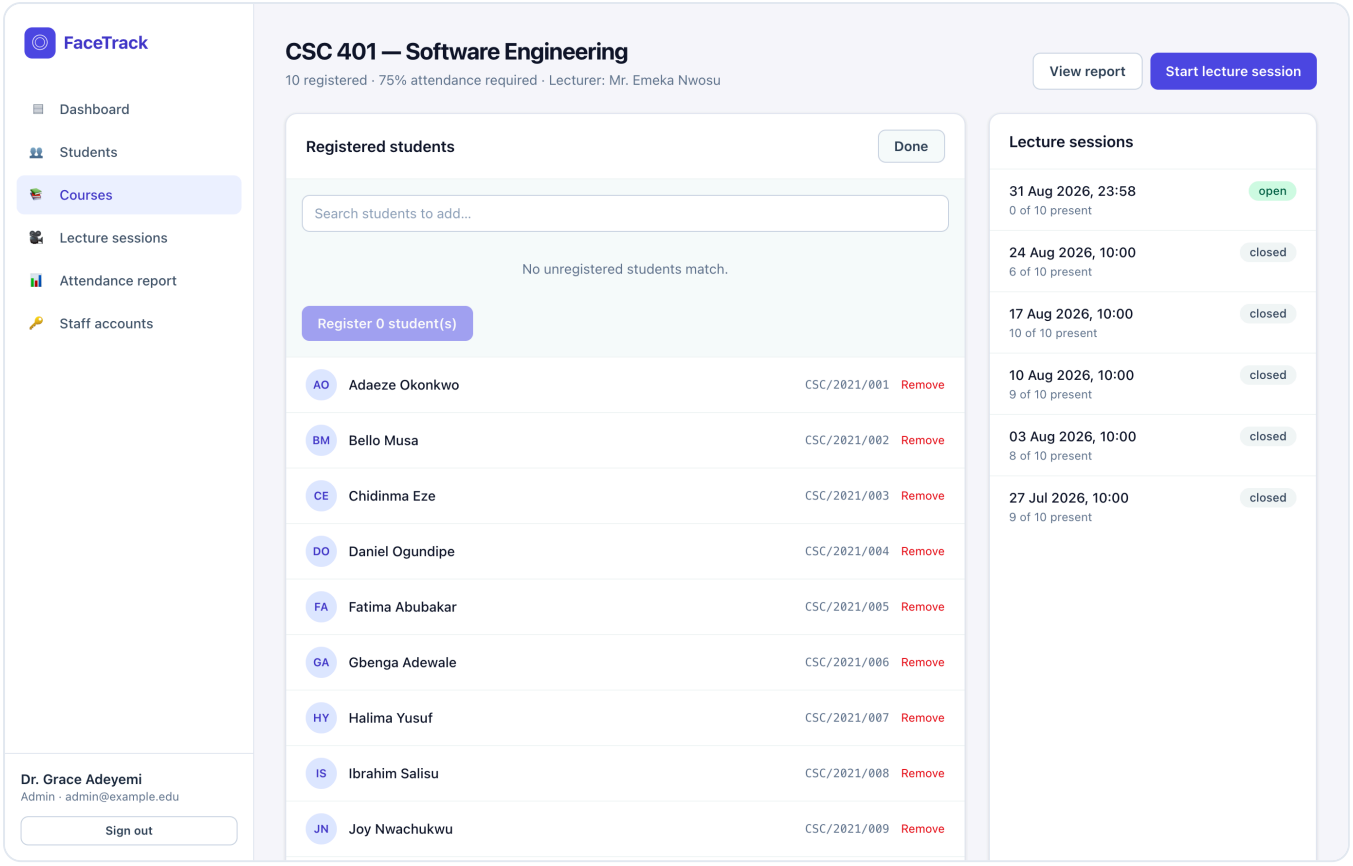


Figure 7. The class list with the add-students picker open. On the right, every session held for this course with its attendance count.

4.8 Lecture sessions EVERYONE `/sessions`

A **session** is one sitting of one lecture. Attendance can only be recorded into an open session.

1. Choose the **course** from the dropdown — it shows how many students are registered on each.
2. Type the **venue** (for example *LT1*) so the register is identifiable later.
3. Set **Late after (mins)**. Anyone recognised after this many minutes is marked *late* instead of *present*. Default 15.
4. Press **Start session & open camera**.

The table below lists every session. Open ones offer **Capture** and **Close**; closed ones offer **View register**.

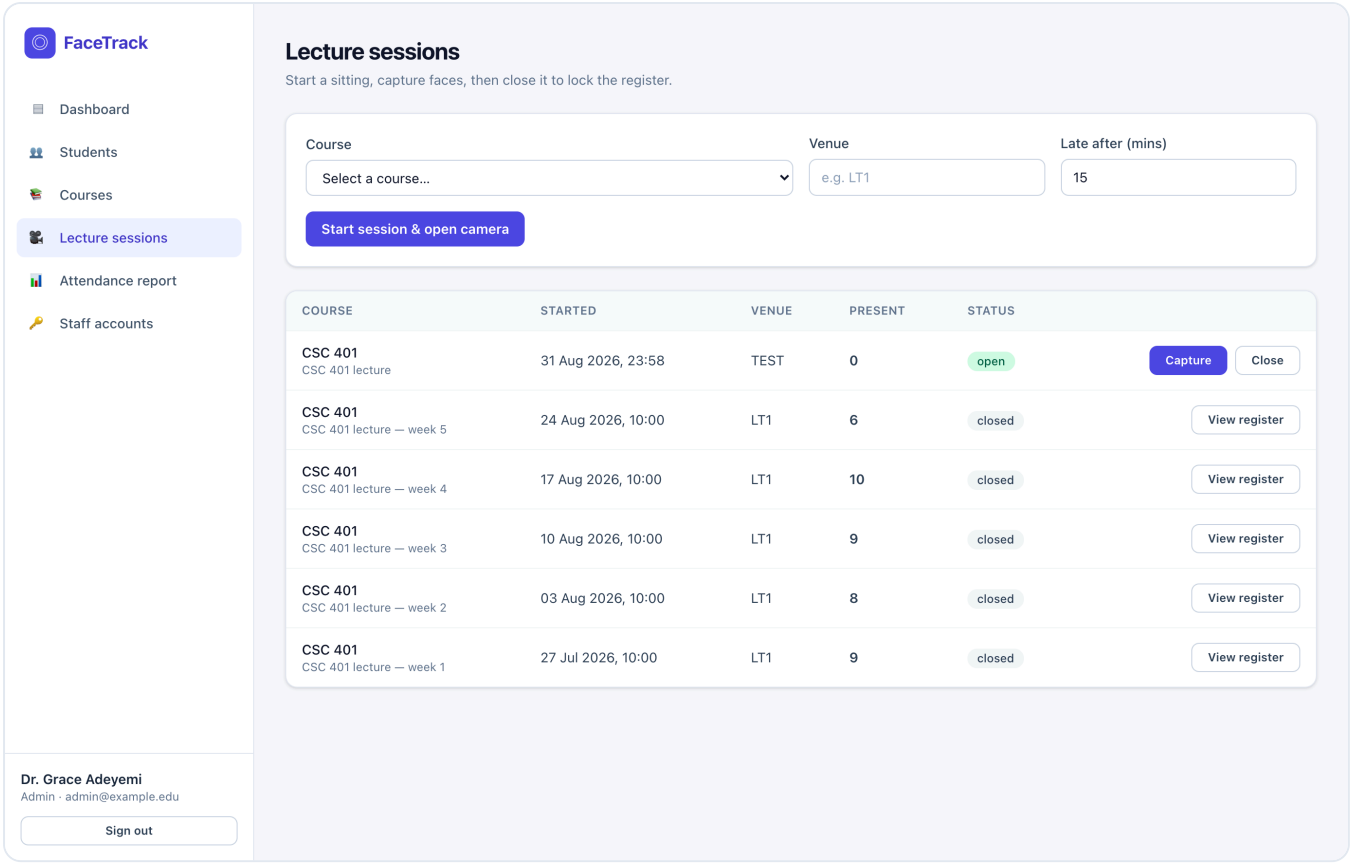


Figure 8. Starting a session, above the history of every sitting held.

> **One open session per course**
> If a session is already open for that course, the system takes you to it rather than creating a duplicate — so a face can never be captured into the wrong sitting.

4.9 Live capture — taking attendance EVERYONE `/sessions/[id]/capture`

The screen you will spend your lecture on.

1. Wait for the button to read **Start capture** rather than *Loading face detector…*, then press it and allow camera access.
2. A red **SCANNING** badge appears. The system now analyses a frame roughly every second.
3. Point the camera at the students. Each detected face gets a coloured box and a name label.
4. Watch the right-hand column: **Present** counts up, students move from **Not yet marked** to **Marked present** as they are recognised.
5. For anyone the camera cannot pick up, press **Mark manually** beside their name.
6. When the lecture ends press **Close session** to lock the register.

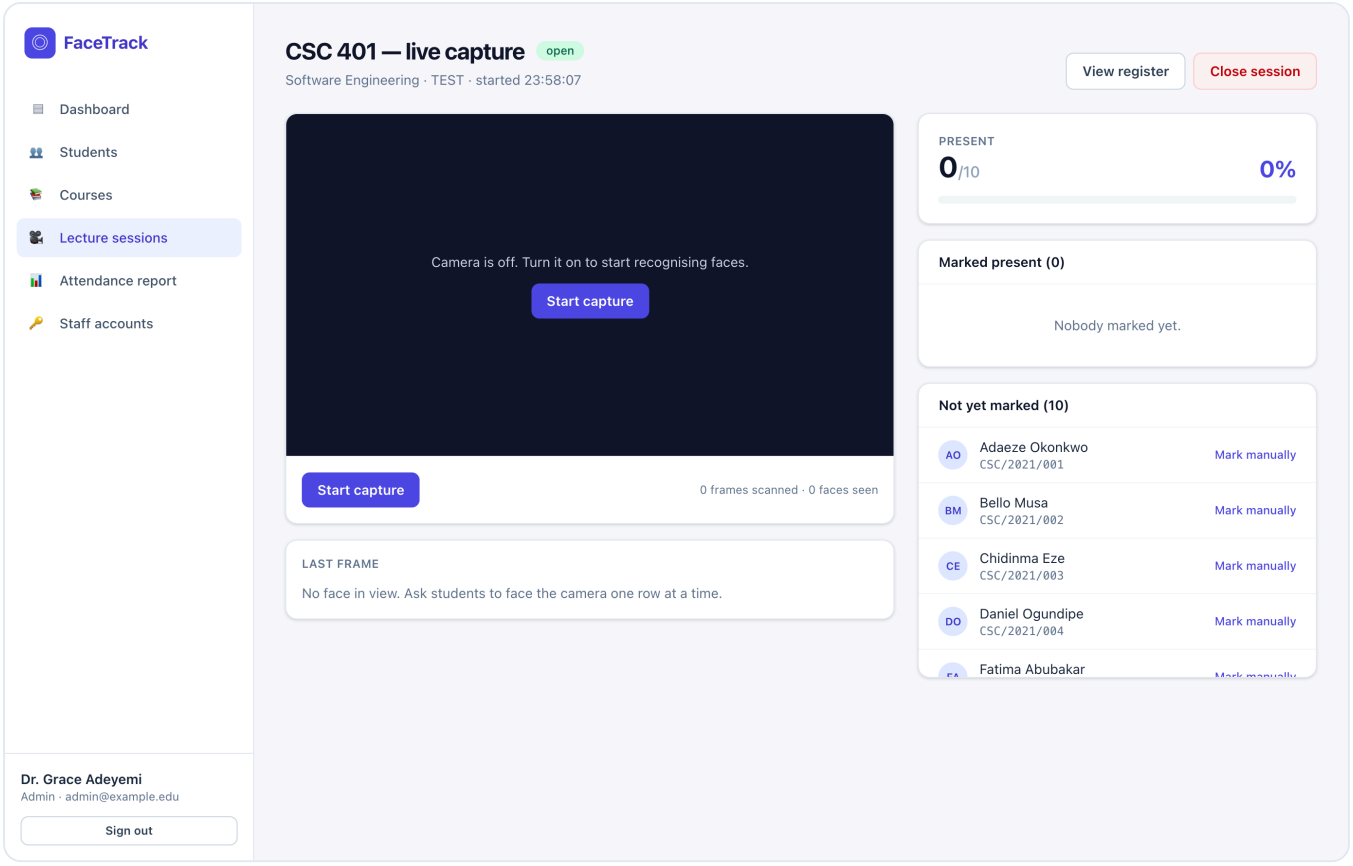


Figure 9. The capture screen before the camera is switched on. Video on the left, the live register on the right, and a **Last frame** panel underneath explaining what the system just decided.

What the box colours mean

Green — newly marked
Recognised and just saved to the register. The count on the right goes up.

Blue — already marked
Recognised, but already recorded for this session. Nothing changes — safe to stay in frame.

Red — not recognised
Either not registered on this course, or has no face template enrolled.

Getting the best results in a hall

Faces need to be reasonably large in the frame, so one camera at the front will not reach the back row. The most reliable arrangement is to have students walk past the camera as they come in, or to pan the camera along each row in turn while you set up.

The frames counter

"N frames scanned · N faces seen" under the video tells you the system is genuinely working even when nobody is in view. If frames are not climbing, capture is paused.

4.10 Session register EVERYONE

`/sessions/[id]`

The formal record of one lecture — who was present, who was late, who was absent.

- **Present** table — each student with the time they were marked, their status, and the method. Face records also show the match distance in brackets: lower means a stronger match.

- **Absent** panel — everyone on the class list who was never marked.
- **Export CSV** — downloads the register, absentees included, for your records.
- **Remove** — deletes a single record if it was captured in error.
- **Close / Reopen session** — locks or unlocks the register.

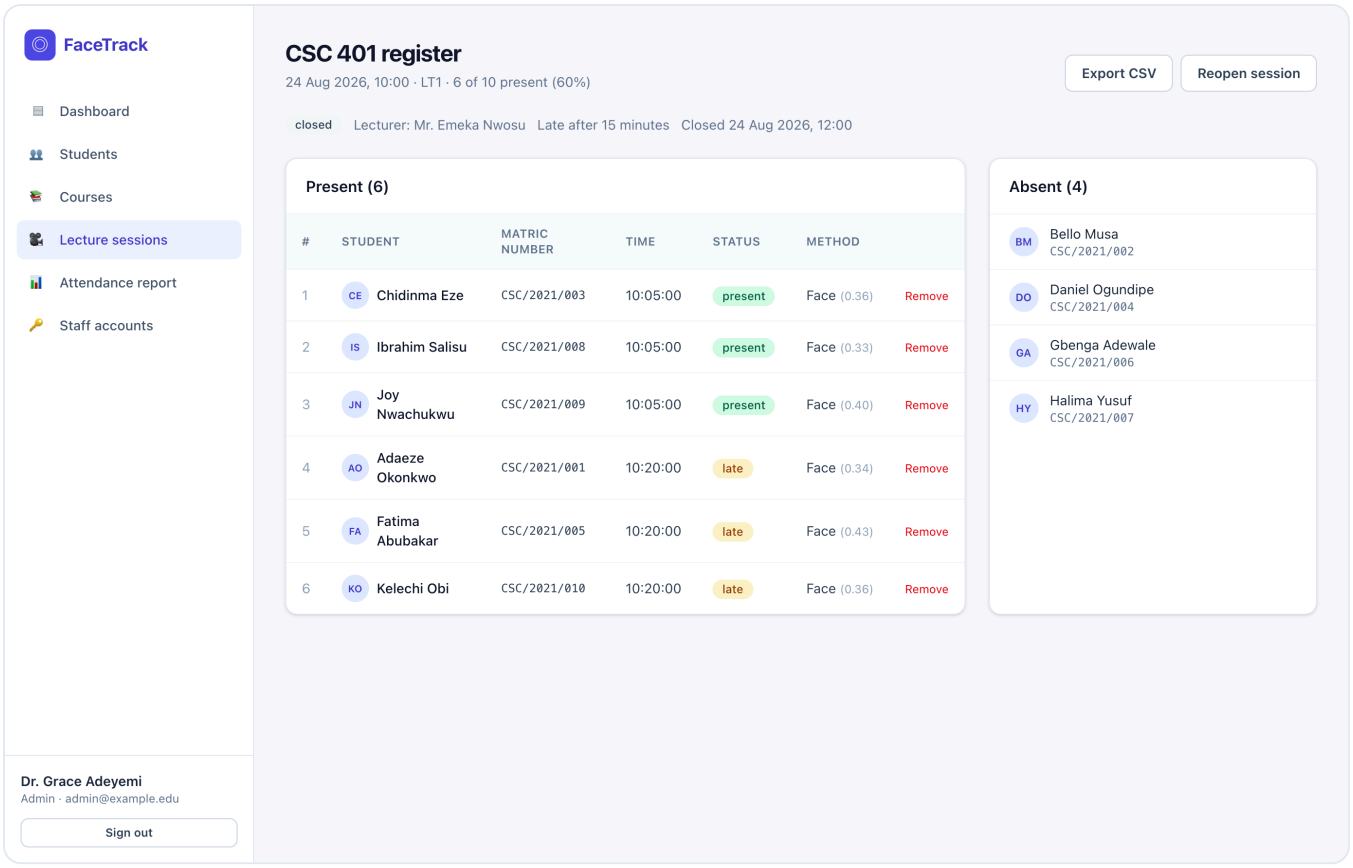


Figure 10. A closed register. The figure in brackets beside *face* is the match distance — the numerical strength of the recognition.

4.11 Attendance report EVERYONE `/reports`

The whole semester for one course on a single grid — this is the screen to show during a demonstration or defence.

1. Pick a **course** from the dropdown.
2. Read the four summary cards: sessions held, students registered, average attendance, and how many fall below the requirement.
3. Read the grid — one row per student, one column per lecture:
   - **P** (green) — present · **L** (amber) — late · **A** (red) — absent
4. The **%** column and the **Eligibility** badge show whether each student meets the course requirement.
5. Press **Export CSV** for a spreadsheet copy of the entire grid.

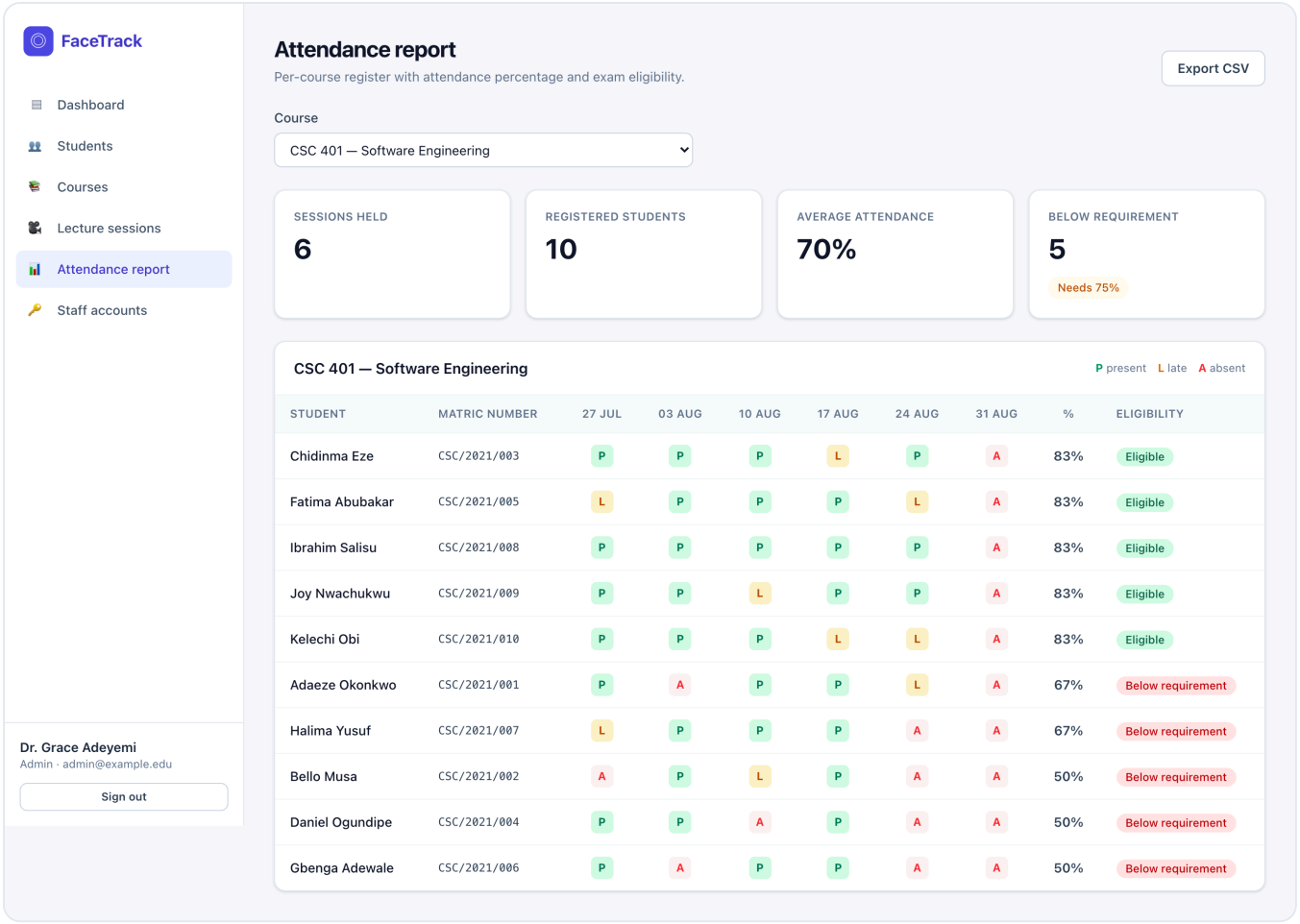


Figure 11. The attendance report. Rows are sorted best-attendance first, so students at risk of being barred from the exam collect at the bottom.

4.12 Staff accounts ADMIN ONLY

`/staff`

Where lecturers and administrators are created. This item does not appear in the sidebar for lecturers.

1. Fill in **full name**, **email** and a **password** of at least six characters.
2. Choose the **role** — *Lecturer* for teaching staff, *Administrator* for full access.
3. Optionally add a staff ID and department.
4. Press **Create account**. The new account appears in the list on the right and can sign in immediately.

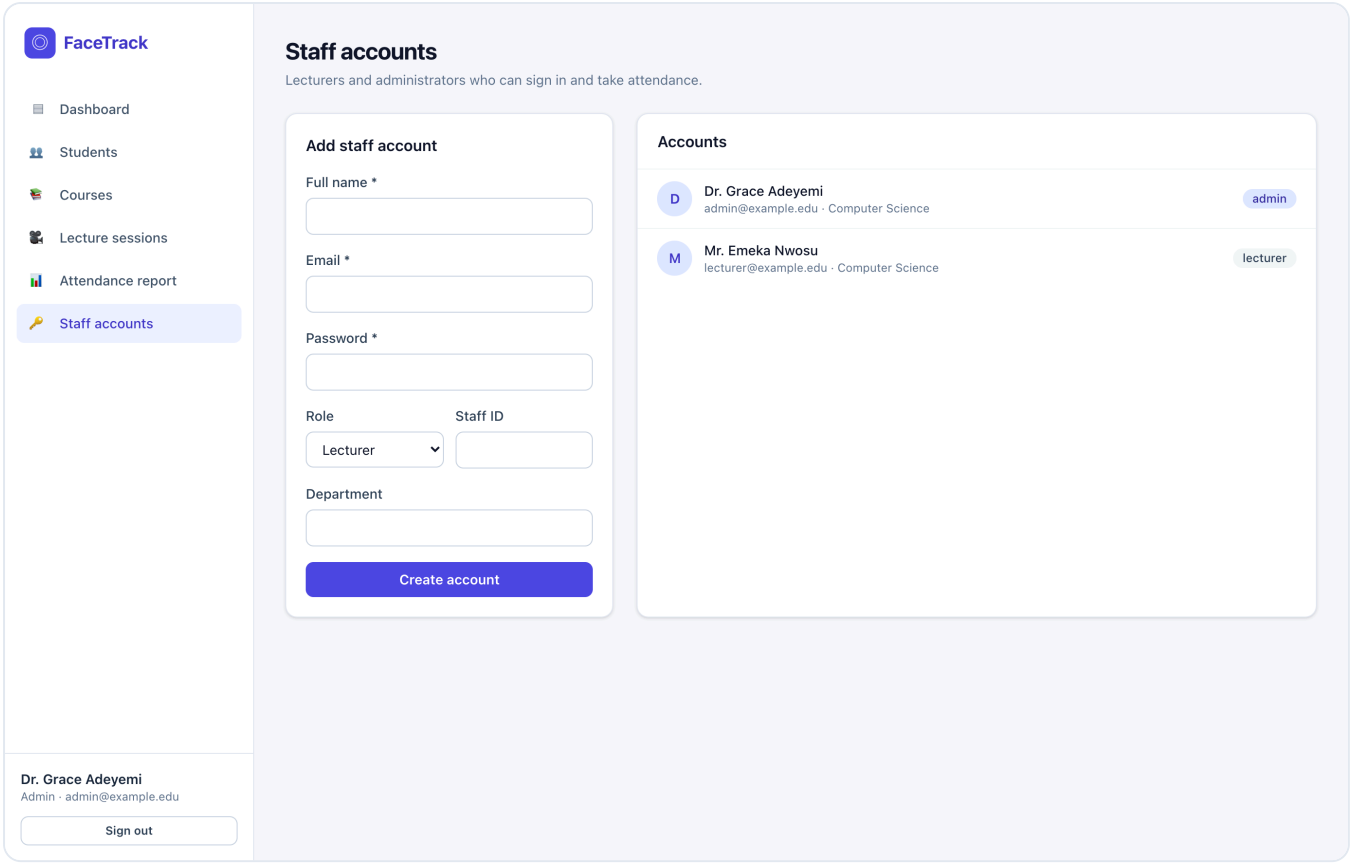


Figure 12. Staff account management. Create a lecturer here before assigning them a course in section 4.6.

5. User flows

The three journeys that string the screens together.

ADMINISTRATOR Flow A — Setting up a course for the first time

STEP 1 Sign in — as administrator → **STEP 2** Register students — details + 3 face samples each → **STEP 3** Create course — code, title, lecturer, 75% → **STEP 4** Build class list — add students to the course

Do this once per course, before the first lecture. Step 2 is the slow one — budget about a minute per student, and do it in a well-lit room.

LECTURER Flow B — Taking attendance in a lecture

STEP 1 Start session — course, venue, late cut-off → **STEP 2** Start capture — allow the camera → **STEP 3** Scan the room — green boxes = marked → **STEP 4** Mop up — mark stragglers manually → **STEP 5** Close session — locks the register

This is the routine you repeat every lecture. Start the session a minute early so the models finish loading, and always close it at the end — a session left open can still accept attendance.

LECTURER / ADMIN Flow C — End-of-semester reporting

STEP 1 Open reports — choose the course → **STEP 2** Check the grid — P / L / A per lecture → **STEP 3** Review eligibility — who is below 75% → **STEP 4** Export CSV — file the record

A worked example, end to end

Registering one student and marking them present, from a standing start:

#	WHERE	WHAT YOU DO
1	Sign in	Log in as `admin@example.edu` / `admin1234`
2	Students → Register student	Enter *CSC/2021/011* and the student's name
3	↳ Face enrolment	Turn on the camera, capture 3 samples, **Save student**
4	Courses → CSC 401	**+ Add students**, tick the new student, **Register 1 student**
5	↳ same page	**Start lecture session**
6	Live capture	**Start capture**, sit the student in front of the camera
7	↳ watch	Green box appears with their name; **Present 1/11**
8	↳ finish	**Close session**, then check **Attendance report**

6. Use cases

Ten documented scenarios the platform supports.

UC-01 Staff authentication

Administrator / Lecturer

Goal	Gain access to the attendance platform.
Steps	Open the site → enter email and password → press Sign in.
Outcome	A signed session cookie is issued, valid for 8 hours; the dashboard opens.
If it fails	A generic “Invalid email or password” message is shown — the system never reveals which field was wrong.

UC-02 Enrol a student's face

Administrator

Goal	Create the biometric template that makes a student recognisable.
Before	The student is physically present, in good light.
Steps	Students → Register student → enter details → turn on camera → capture 3 samples → Save.
Outcome	Three 128-number descriptors are stored. The student shows a green *3 samples* badge.
Rules	Saving is refused without at least one valid sample, and matriculation numbers must be unique.

UC-03 Register students on a course

Administrator / Lecturer

Goal	Define who belongs in a class, and therefore who can be recognised in its lectures.
Steps	Courses → open the course → + Add students → tick names → Register.
Outcome	Those students become the recognition candidate set for every session of that course.

UC-04 Start a lecture session

Lecturer

Goal	Open a window during which attendance may be recorded.
Steps	Lecture sessions → choose course, venue and late cut-off → Start session.
Outcome	A session with status *open* is created and the capture screen opens.
Rules	A course with no registered students cannot start a session. Only one session per course may be open at a time.

UC-05 Record attendance by face recognition

Lecturer · primary use case

Goal	Mark students present without a roll call.
Steps	Start capture → point the camera at students → watch names appear.
Outcome	For each recognised face an attendance record is written, timed and stamped *present* or *late*.
Rules	Only students on the course are considered; only within the acceptance threshold; only while the session is open; and only one record per student per session.

UC-06 Mark a student manually

Lecturer

Goal	Record a student the camera cannot recognise — poor light, no template, a camera failure.
Steps	On the capture screen, find them under *Not yet marked* → press Mark manually.
Outcome	A record tagged *manual*, so it stays distinguishable from a biometric one in every report.

UC-07 Close a session

Lecturer

Goal	Freeze the register when the lecture ends.
Steps	Press Close session on the capture screen or the session list.
Outcome	Status becomes *closed* and any further recognition attempt is rejected.

UC-08 Correct a wrong record

Lecturer / Administrator

Goal	Remove an entry captured in error.
Steps	Open the session register → press Remove beside the record → confirm.
Outcome	The record is deleted and the student returns to the absent list.

UC-09 Produce an attendance report

Lecturer / Administrator

Goal	Determine each student's attendance percentage and exam eligibility.
Steps	Attendance report → choose the course → read the grid → Export CSV.
Outcome	A full register with per-student percentages and an eligibility verdict against the course requirement.

UC-10 Create a staff account

Administrator

Goal	Give a new lecturer access to the platform.
Steps	Staff accounts → fill in name, email, password and role → Create account.
Outcome	The account can sign in immediately and be assigned as a course lecturer.

7. Troubleshooting

The problems you are most likely to hit, and what to do about them.

SYMPTOM	WHY IT HAPPENS	WHAT TO DO
The camera button stays greyed out	The face models (~12 MB) are still downloading.	Wait 10–40 seconds. If it never finishes, reload the page — a partial download is retried from scratch.
"Camera permission was denied"	The browser blocked access, usually from an earlier refusal.	Click the padlock in the address bar, set Camera to *Allow*, then reload.
Nothing happens when you turn the camera on	The page is on an insecure address such as a LAN IP.	Use the HTTPS live link, or `localhost` if running it yourself. See section 2.3.
"No face detected" during enrolment	Backlighting, or the student is too far away.	Face the student towards the light, move to about an arm's length, and stay still while capturing.
A red box — a real student is not recognised	They have no face template, or are not on this course's class list.	Check the Students page for a red *Not enrolled* badge, and the course class list. Mark them manually for now, and re-enrol afterwards.
Recognised, but marked *late*	They were recognised after the late cut-off for the session.	Expected behaviour. Set a longer cut-off when starting the session if your lectures fill slowly.
"This session is closed"	Attendance cannot be added to a finished lecture.	Open the session register and press **Reopen session** if it was closed by mistake.
"This course already has an open session"	A previous sitting was never closed.	The system takes you to that session. Close it, then start a new one.
"Register students on this course first"	The class list is empty.	Open the course and add students before starting a session.
The report grid is empty	No sessions have been held for that course yet.	Run at least one lecture session. Only sessions that exist appear as columns.
Attendance stops being recorded mid-lecture	Capture was paused, or the session was closed in another tab.	Check for the red **SCANNING** badge and that the frames counter is climbing.

Known limits of the prototype

Worth stating plainly, especially during a demonstration:

- **Lighting still matters.** In a dim hall the detector finds fewer faces. Manual marking exists for exactly this reason.
- **Range is limited.** Faces must be reasonably large in frame; one camera at the front will not reach the back of a 300-seat hall.
- **There is no liveness detection.** A printed photograph held to the camera would pass. Defeating that needs blink or depth sensing, which is outside this prototype's scope.
- **The threshold is a trade-off, not a certainty.** Stricter settings reject more genuine students; looser settings risk matching the wrong one.
- **Matching scales linearly** with class size — fine for tens or low hundreds of students per course.

8. Quick reference card

Keep this page open during a demonstration.

Sign in

Live link	`https://attendace-system-david.vercel.app/`	
Administrator	`admin@example.edu`	`admin1234`
Lecturer	`lecturer@example.edu`	`lecturer1234`

Run a lecture — 6 steps

1. Lecture sessions → pick the course
2. Set venue and late cut-off → Start session
3. Wait for **Start capture** to go live → press it
4. Point the camera; green boxes = marked
5. Mark stragglers manually
6. **Close session**

Enrol a face — 4 steps

1. Students → + Register student
2. Matric number and full name
3. Turn on camera → capture 3 samples
   straight · slight left · slight right
4. Save student

Box colours

Newly marked	
Already marked	
Not recognised	

Report symbols

P present · **L** late · **A** absent — the **%** column is measured against the course requirement (75% by default).

Before real use

Replace the demonstration accounts above with strong passwords under **Staff accounts**, and remember that students' face templates are personal biometric data — enrol only with their knowledge and consent.